Ex4a: Information Retrieval using NLTK

1. Learning Objectives

- To understand the basic concept of Information Retrieval (IR).
- To preprocess text using NLTK.
- To perform lowercasing, punctuation removal, tokenization, and stop-word removal.
- To convert text documents into numerical vectors using TF-IDF.
- To calculate cosine similarity between a user query and reviews.
- To retrieve the top-k most relevant reviews.
- To understand how text preprocessing improves information retrieval.

2. Methods Used / Python Libraries Involved

Library / Method	Purpose	Reason to Use
Pandas	Load and manipulate the dataset	Easy handling of CSV data and DataFrames
NLTK	Natural Language Processing	Provides tokenization and stop-word removal
word_tokenize()	Splits text into individual words	Required for text preprocessing
stopwords	Provides common English stopwords	Provides common English stopwords
TfidfVectorizer	Converts text into TF-IDF vectors	Represents text numerically based on word importance

3. Program Code

```
import pandas as pd import re import nltk from nltk.corpus
import stopwords from nltk.tokenize import word_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

```
 nltk.download('punkt')
nltk.download('punkt_tab'
)
nltk.download('stopwords'
)


df = pd.read_csv("Reviews.csv")
df    =    df[['Text']]    df    =
df.dropna() df = df.head(10000)
 print("Number    of    reviews:",    len(df))
stop_words = set(stopwords.words('english'))
 def  preprocess_text(text):  text  =  text.lower()  text  =
re.sub(r'[^a-zA-Z\s]', '', text) words = word_tokenize(text)
words = [word for word in words if word not in stop_words]
return ' '.join(words)


df['Cleaned_Text']    =    df['Text'].apply(preprocess_text)
vectorizer    =    TfidfVectorizer()    tfidf_matrix    =
vectorizer.fit_transform(df['Cleaned_Text']) print("TF-IDF
Matrix Shape:", tfidf_matrix.shape)
 def retrieve_reviews(query, top_k=5): cleaned_query =
preprocess_text(query)          query_vector          =
vectorizer.transform([cleaned_query]) similarity_scores
= cosine_similarity( query_vector, Tfidf_matrix
).flatten()  top_indices  =  similarity_scores.argsort()[-
    top_k:][::-1] print("\nQuery:", query)
    print("\nTop", top_k, "Relevant Reviews:\n")
 for i in top_indices:
        print("Similarity                          Score:",
              round(similarity_scores[i], 4))
        print("Original Review:")
        print(df.iloc[i]['Text'])
        print("\nCleaned Review:")
        print(df.iloc[i]['Cleaned_Text'])
        print("-" * 80)


retrieve_reviews("great product with fast shipping", 5)
retrieve_reviews("disappointed", 5)
```

4. Output

```
Number of reviews: 20
TF-IDF Matrix Shape: (20, 44)

Query: great product with fast shipping

Top 5 Relevant Reviews:

Similarity Score: 0.8538
Original Review:
This is a great product and it arrived with fast shipping.

Cleaned Review:
great product arrived fast shipping

--------------------------------------------------------------------------------------------------------------------------------
Similarity Score: 0.5296
Original Review:
The shipping was fast and the product was exactly as described.

Cleaned Review:
shipping fast product exactly described

--------------------------------------------------------------------------------------------------------------------------------
Similarity Score: 0.5055
Original Review:
Great product, excellent quality and very fast delivery.

Cleaned Review:
great product excellent quality fast delivery

--------------------------------------------------------------------------------------------------------------------------------
Similarity Score: 0.415
Original Review:
Fast shipping and good packaging. I am happy with the purchase.

Cleaned Review:
fast shipping good packaging happy purchase

--------------------------------------------------------------------------------------------------------------------------------
Similarity Score: 0.3138
Original Review:
The delivery was fast but the product was average.
```

5. Learning Outcome

- I learned how to build a basic Information Retrieval system using Python.
- Learned how to preprocess review text using NLTK.
- Understood the importance of tokenization and stop-word removal.
- Learned how TF-IDF represents text as numerical vectors.
- Learned how cosine similarity measures the similarity between a query and documents.
- Successfully retrieved the top-k relevant Amazon product reviews.
- Understood how similarity scores can be used to rank search results.
- Gained practical knowledge of applying NLP techniques to real-world text data.